**SOKOINE UNIVERSITY OF AGRICULTURE**

School of Engineering and Technology | Department of Agricultural Engineering

# AEN 214: INTRODUCTION TO ELECTRONICS

## OFFICIAL COURSE SYLLABUS & LEARNING CONTRACT

Instructor: Dr. Kadeghe Goodluck Fue, PhD, P.Eng (T) | Laboratory: Electronics & Precision Agriculture Lab
Course Status: Elective | Credit Rating: 6 Credits | Total Notional Hours: 60 Hours | Academic Year: 2025/2026

## 1. Course Aim & Rationale

This course serves as a rigorous, hands-on introduction to the fundamental principles, physics, and applications of analog and digital electronics within the domain of agricultural and biosystems engineering. Modern agricultural transformation—ranging from smart drip irrigation automation and climate-controlled greenhouses to precision livestock telemetry, autonomous tractors, and grain quality monitoring—relies on solid-state transducers, discrete analog amplifiers, power semiconductor switching actuators, and digital processing logic. AEN 214 bridges basic electro-technics with modern AgriTech cyber-physical hardware systems.

## 2. Course Expected Learning Outcomes (CELOs)

By the completion of this course, engineering students will be able to:

1. Describe and evaluate fundamental concepts of analog and digital electronics and articulate their strategic importance in modern agricultural mechanization and automation.
2. Analyze electronic components and determine their electrical characteristics, including linear/non-linear resistors, capacitors, inductors, diodes, and transistors under DC, transient, and high-frequency conditions.
3. Analyze, simulate, and design basic analog circuits—including multi-stage amplifiers, active/passive filters, DC voltage regulators, and timing oscillators—tailored for agricultural sensor conditioning.
4. Synthesize and minimize digital logic systems, utilizing Boolean algebraic laws, De Morgan's theorems, and Karnaugh maps to implement combinational and sequential logic for farm process automation.
5. Explain principles of data conversion, analyzing Analog-to-Digital Converters (ADCs) and Digital-to-Analog Converters (DACs) interfacing physical farm sensors with digital processors.
6. Apply microcontrollers and power semiconductors (Power BJTs, MOSFETs) to drive inductive farm actuators (solenoid valves, motor drivers, relays, and injection injectors).
7. Demonstrate hands-on laboratory mastery in circuit prototyping, breadboarding, schematic design, multimeter and oscilloscope diagnostics, and fault troubleshooting.
8. Evaluate emerging electronics technologies in agricultural engineering, including IoT sensor nodes, solar photovoltaic balance-of-system electronics, and autonomous agricultural robotics.

## 3. Detailed Course Content & Lecture Modules

| Module # | Curriculum Topic | Key Subtopics & Engineering Concepts | Hours Allocated |
|---|---|---|---|
| Lecture 1 | Passive Electronic Components | Resistors (types, color codes, power rating, TCR, thermistors, LDRs, strain gauges); Capacitors (electrostatics, dielectrics, codes, transients); Inductors; Reactance (Xc, Xl) and resonance. | 6.0 Hours |
| Lecture 2 | Semiconductor Physics | Bohr atomic model; Energy bands (VB, CB, Eg); Intrinsic vs extrinsic; Doping (P-type, N-type); Carrier transport; PN junction depletion layer; Built-in barrier V0; Hall effect sensor operation. | 6.0 Hours |
| Lecture 3 | Semiconductor Diode & Circuits | Shockley equation; Dynamic resistance rd derivation; Piecewise approximations; Half-wave and full-wave bridge rectifiers; Smoothing filters and ripple factor; Diode clippers and clampers. | 7.0 Hours |
| Lecture 4 | Special Purpose Diodes | Zener diodes (Zener vs Avalanche breakdown); Shunt voltage regulator design; LEDs (direct bandgap, electroluminescence, PAR grow lighting); Schottky barrier diodes (solar bypass, fast recovery); Photodiodes. | 6.0 Hours |
| Lecture 5 | Bipolar Junction Transistor (BJT) | BJT physics (NPN, PNP); Biasing rules; CB, CE, CC configurations; Current gains (α, β); DC load line and Q-point; Thermal runaway; Low-side saturated switch; CE small-signal amplifier. | 8.0 Hours |
| Lecture 6 | Field Effect Transistor (FET) | JFET operating principles; Pinch-off VP; Shockley transfer equation; Transconductance gm; DE-MOSFET and E-MOSFET; Power MOSFETs (IRF540N) in solar pump and motor drivers; ESD handling. | 7.0 Hours |
| Lecture 7 | Number Systems & Arithmetic | Positional radices (Binary, Octal, Decimal, Hexadecimal); Inter-base conversion algorithms; Signed two's complement arithmetic and overflow; Digital codes (BCD, Excess-3, Gray code); ASCII & NMEA. | 6.0 Hours |
| Lecture 8 | Introduction to Digital Electronics | Logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR); Universal gate synthesis; Propagation delays; Half and Full adders; Combinational greenhouse interlocks; Latches and Flip-Flops (SR, D, JK). | 7.0 Hours |
| Lecture 9 | Boolean Algebra & K-Maps | Boolean axioms and algebraic laws; De Morgan's theorems; Canonical SOP (minterms) and POS (maxterms); 2, 3, and 4-variable Karnaugh Maps (Gray code order); Don't Care (X) optimization; Grain dryer case study. | 7.0 Hours |

## 4. Teaching and Learning Methodology

i. Interactive Lectures & Discussions: Interactive lectures covering passive components, semiconductor physics, and digital logic with live mathematical derivations and circuit simulations.
ii. Hands-on Laboratory Experiments: Practical bench sessions in the Electronics and Precision Agriculture Lab: breadboard circuit assembly, oscilloscope waveform measurement, rectifier diode testing, BJT/MOSFET switching, and digital logic gate verification.
iii. Guest Lectures & Industry Insights: Guest lectures from agricultural automation and sensor systems specialists covering tractor electronic control units (ECUs) and solar irrigation installations.
iv. Self-Directed Problem Sets & Open Coursework: Tiered engineering problem sets and design exercises reinforcing analytical calculations, component selection, and safety derating in preparation for theory assessments.

## 5. Assessment Methods & Grading Policy

| Assessment Component | Scope / Topic Coverage | Weighting (%) | Format / Deliverable |
|---|---|---|---|
| Theory Test 1 | Topics 1 through 4 (Passive Components, Semiconductor Physics, Diodes & Special Diodes) | 20% | Closed-book written engineering theory examination |
| Theory Test 2 | Topics 5 through 9 (BJTs, FETs/MOSFETs, Number Systems, Digital Logic & Boolean Algebra) | 20% | Closed-book written engineering theory examination |
| Continuous Assessment (CA) Subtotal | Two Theory Tests (Test 1 + Test 2 Combined) | 40% | Cumulative Continuous Assessment (CA) component |
| University Examination (UE) | Comprehensive Course Coverage (Topics 1 through 9) | 60% | Formal 3-hour theoretical and design university examination |
| Total Final Grade | Theory Tests (40%) + University Examination (60%) | 100% | SUA Grading Scale (A: 70%+, B+: 60-69%, B: 50-59%, C: 40-49%) |

## 6. Prescribed Textbooks & Reading List

1. Boylestad, R. L., & Nashelsky, L. (2018). Electronic Devices and Circuit Theory (11th ed.). Pearson Education.
2. Sedra, A. S., & Smith, K. C. (2020). Microelectronic Circuits (8th ed.). Oxford University Press.
3. Floyd, T. L. (2017). Electronic Devices: Conventional Current Version (10th ed.). Pearson.
4. Mano, M. M., & Ciletti, M. D. (2017). Digital Design: With an Introduction to Verilog HDL (6th ed.). Pearson.
5. Karim, M. A., & Chen, X. (2017). Digital Design: Basic Concepts and Principles. CRC Press.
6. Person (2015). Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering (6th ed.). Pearson. ISBN-10: 9781292076683.
7. Johnson, M. A., & Smith, K. L. (2023). Fundamentals of Electronics: An Introduction. IEEE Transactions on Education, 66(3), 456-465.
8. Brown, R. J., & Davis, P. W. (2023). Basic Electronic Components and Circuits. Journal of Electrical Engineering, 44(2), 78-85.

9. Williams, S. E., & Wilson, T. A. (2023). Introduction to Semiconductor Devices and Circuits. Journal of Applied Physics, 135(6), 065420.